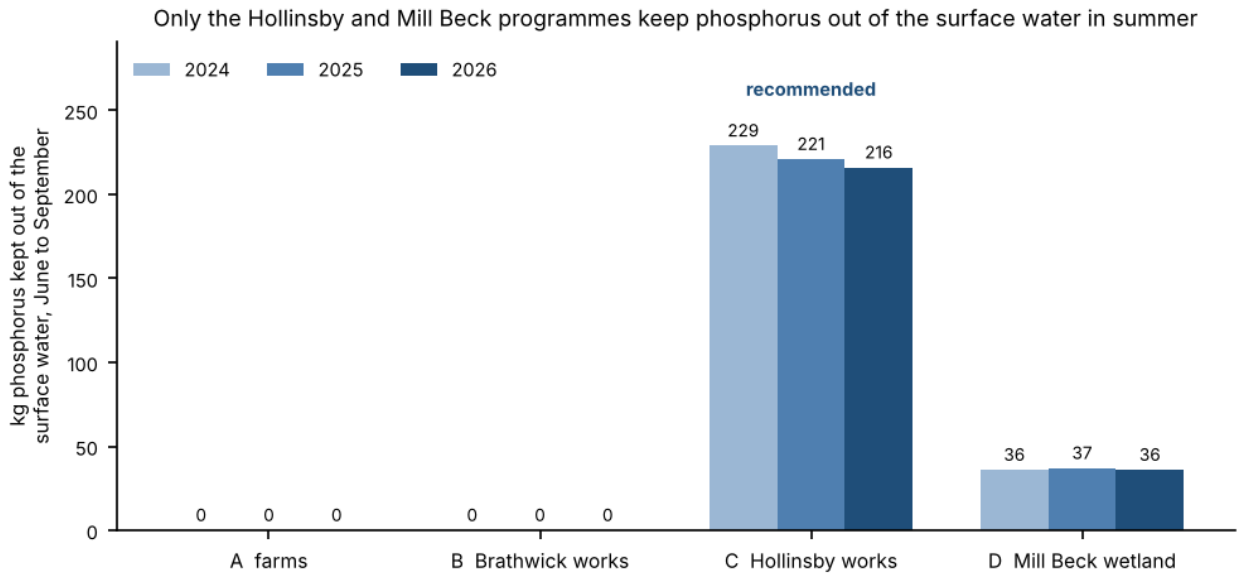
Recovery Fund 2027: the Hollinsby tertiary filter keeps the most phosphorus out of the bloom water

Recommendation: back programme C, tertiary filtration at the Hollinsby works. Under the fund's rules it keeps 222 kg of phosphorus a season out of the lake's surface water. The runner-up is programme D, the Mill Beck outfall wetland, at 36 kg.

Programmes A (Lenning farms) and B (Brathwick works) both act on phosphorus carried by the River Lenning, and the Lenning does not reach the surface water in summer. It is spring-fed and runs colder than the lake's top 5 m on every day from June to September, so its water sinks below the thermocline when it enters the lake. That held on 122 of 122 days in 2024, 122 of 122 in 2025 and 122 of 122 in 2026 (river daily mean against the buoy's 0 to 5 m mean). The buoy's 10 m conductivity rises through each summer with the river's water while the 1 m and 5 m readings stay level. Brathwick's effluent and the farms' runoff therefore add nothing to the surface water before the October overturn, and neither programme keeps any phosphorus out of it in the bloom season.

Hollinsby's effluent is warmer than the surface water in every month from June to September, so it spreads in the water the blooms grow in. Applying the appraisal's credited reduction (71%) to the works' June-to-September loads in Fellmoor Water's returns gives 229, 221 and 216 kg for 2024 to 2026.



Figures by programme (kg a season, June to September)

Programme	2024	2025	2026	Average
A Lenning farm infrastructure programme	0	0	0	0
B Brathwick WwTW chemical phosphorus removal	0	0	0	0
C Hollinsby WwTW tertiary filtration	**229**	**221**	**216**	**222**
D Mill Beck outfall treatment wetland	36	37	36	36

Sources: Fellmoor Water monthly returns to September 2026; Lenning Foot station; Fellside buoy; the trust's Mill Beck outfall monitoring; programme_appraisals_2027.xlsx.

The farm programme would still cut what the Lenning delivers to the deep water and to the lake after the overturn, and the trust should keep it on the list for the next round. It does not change which programme does most for the 2027 blooms.

Workings, the 2023-24 winter inflow, the north basin results and the Connect Right tally are in selwith_phosphorus_workings.xlsx.

Note: figures follow the fund's rules 2.1 to 2.3, the appraisal's credited reduction applied to what each source added to the 0 to 5 m surface water between 1 June and 30 September, the three seasons averaged before rounding to the kilogram.